

Operating Systems

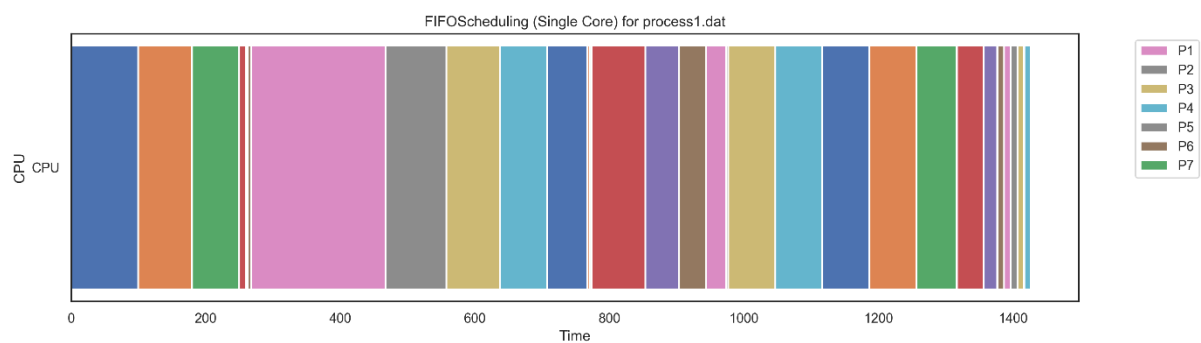
Assignment 3

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September 15, 2025

1.SINGLE CORE PROCESSOR

FIFO



Process1.dat

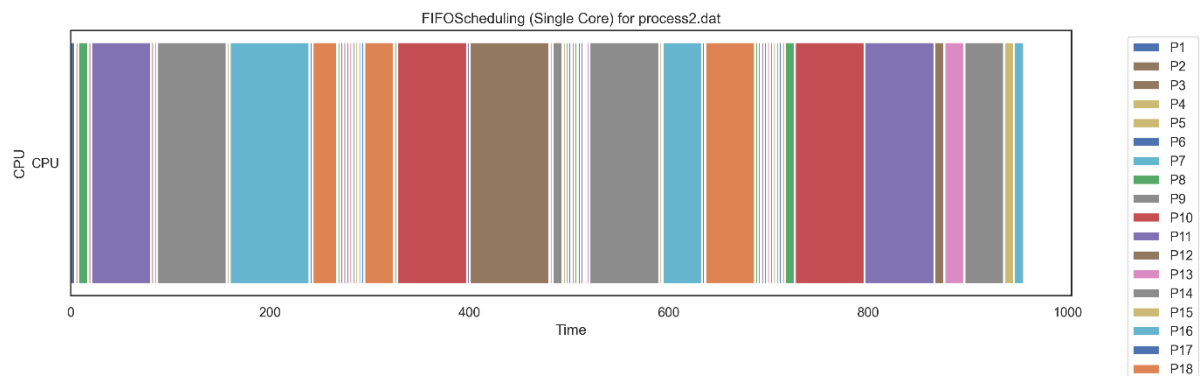
Time to Complete: 1427

Maximum TAT: 1423

Average TAT: 1091

Runtime of Simulator: 1427

Total runtime: 31184 microseconds



Process2.dat

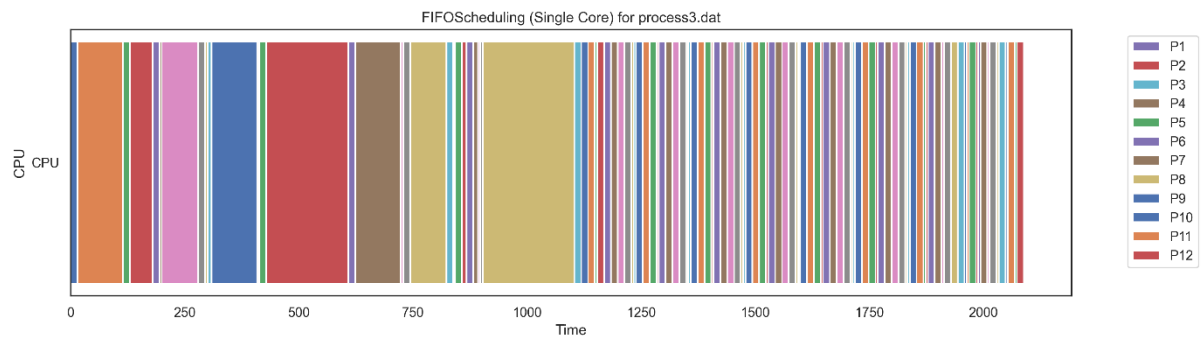
Time to Complete: 957

Maximum TAT: 931

Average TAT: 618.444

Runtime of Simulator: 957

Total runtime: 29147 microseconds



Process3.dat

Time to Complete: 2089

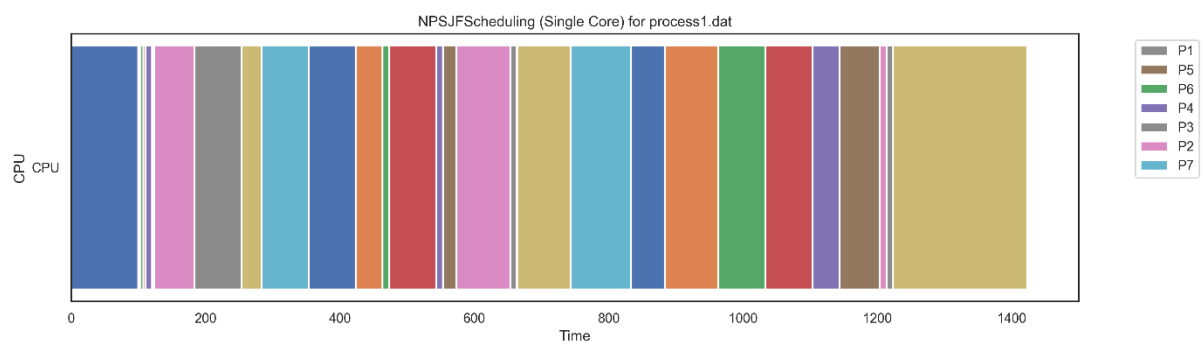
Maximum TAT: 2062

Average TAT: 1249.25

Runtime of Simulator: 2089

Total runtime: 18975 microseconds

NPSJF



Process1.dat

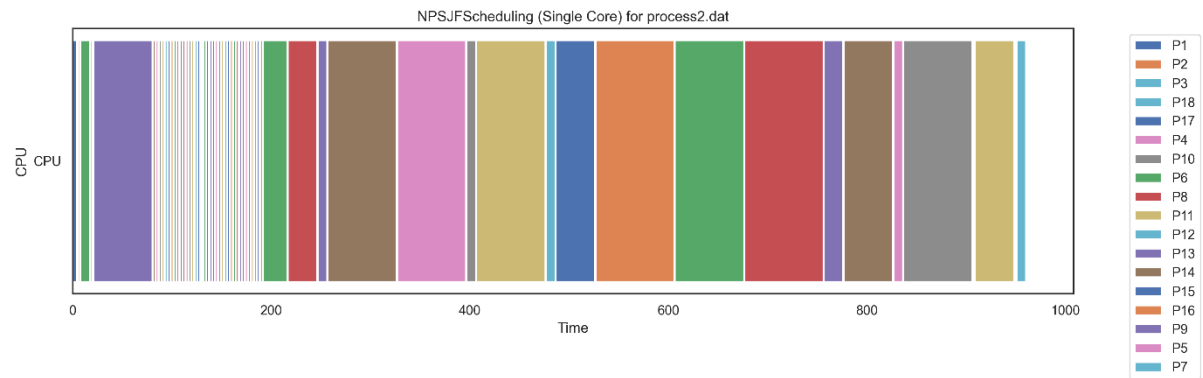
Time to Complete: 1429

Maximum TAT: 1419

Average TAT: 755.286

Runtime of Simulator: 1427

Total runtime: 12319 microseconds



Process2.dat

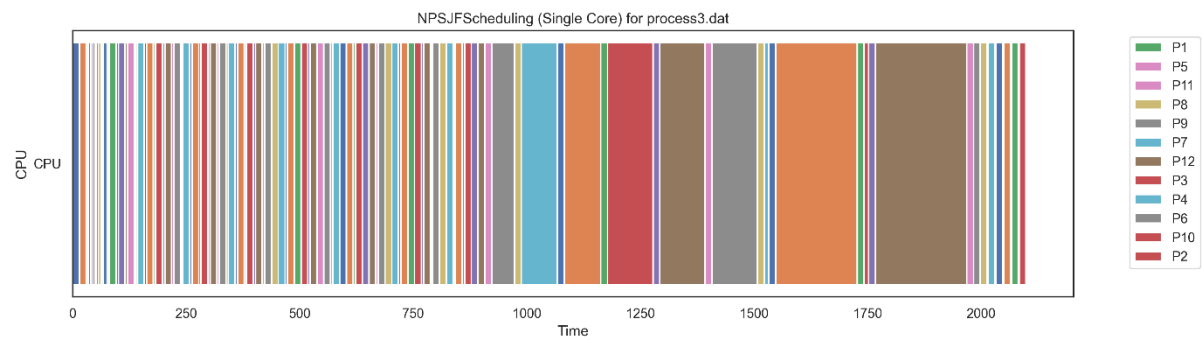
Time to Complete: 961

Maximum TAT: 935

Average TAT: 216.167

Runtime of Simulator: 957

Total runtime: 15967 microseconds



Process3.dat

Time to Complete: 2091

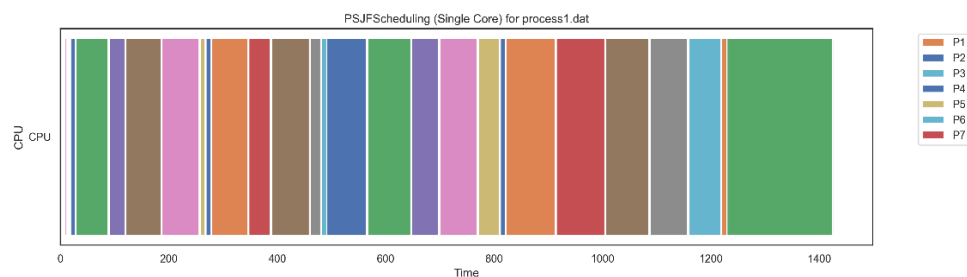
Maximum TAT: 2071

Average TAT: 975.833

Runtime of Simulator: 2089

Total runtime: 30196 microseconds

PSJF



Process1.dat

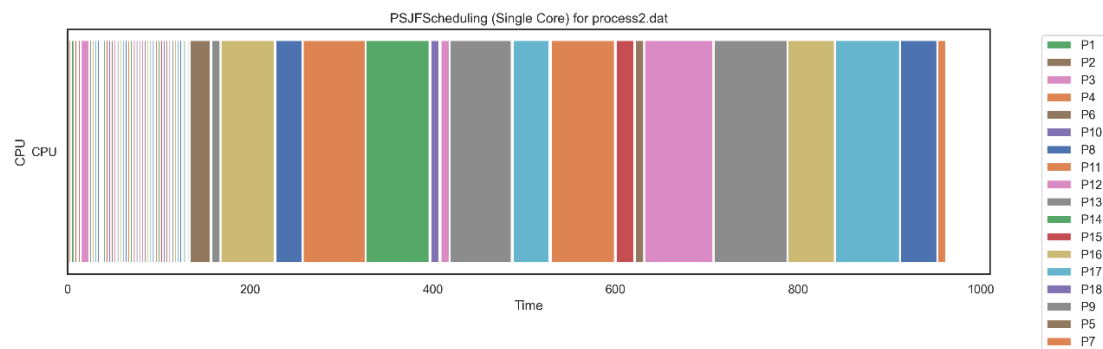
Time to Complete: 1428

Maximum TAT: 1417

Average TAT: 607

Runtime of Simulator: 1428

Total runtime: 19713 microseconds



Process2.dat

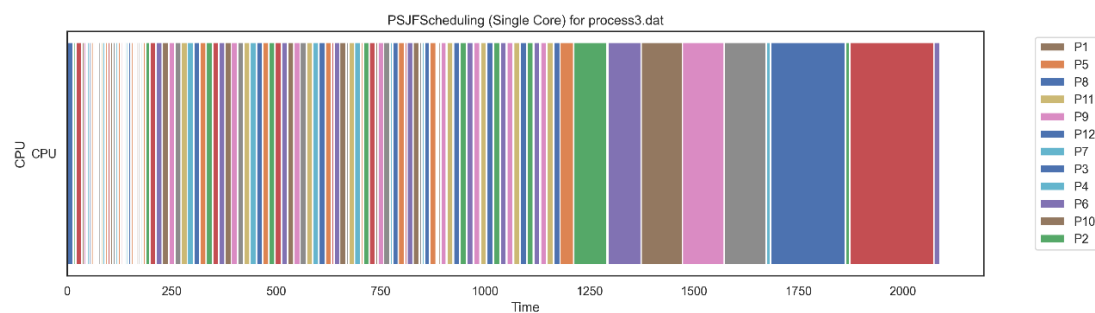
Time to Complete: 963

Maximum TAT: 936

Average TAT: 152.278

Runtime of Simulator: 963

Total runtime: 13417 microseconds



Process3.dat

Time to Complete: 2090

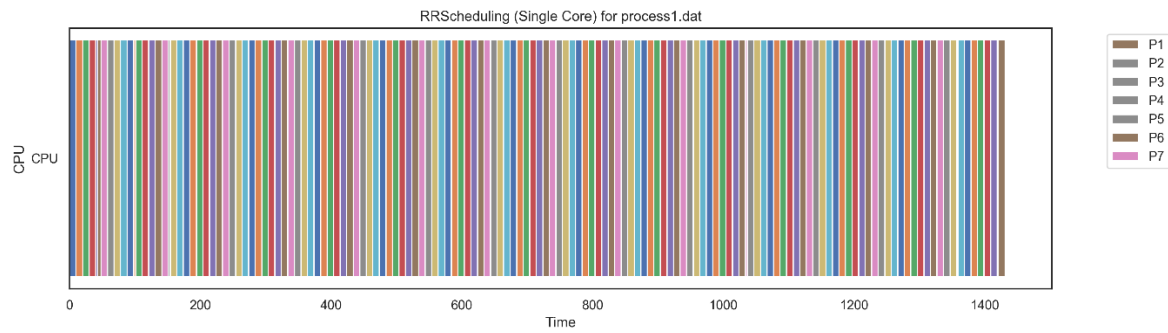
Maximum TAT: 2069

Average TAT: 899.417

Runtime of Simulator: 2090

Total runtime: 32170 microseconds

RR



Process1.dat

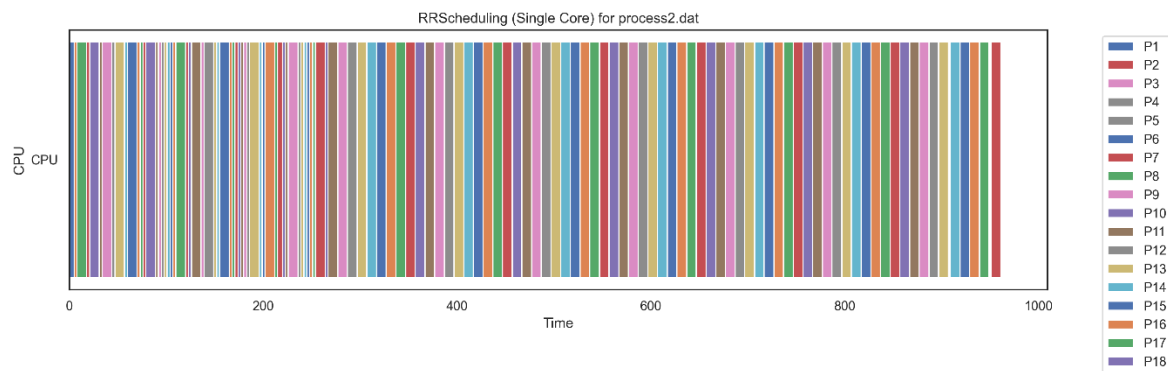
Time to Complete: 1431

Maximum TAT: 1431

Average TAT: 884

Runtime of Simulator: 1429

Total runtime: 29207 microseconds



Process2.dat

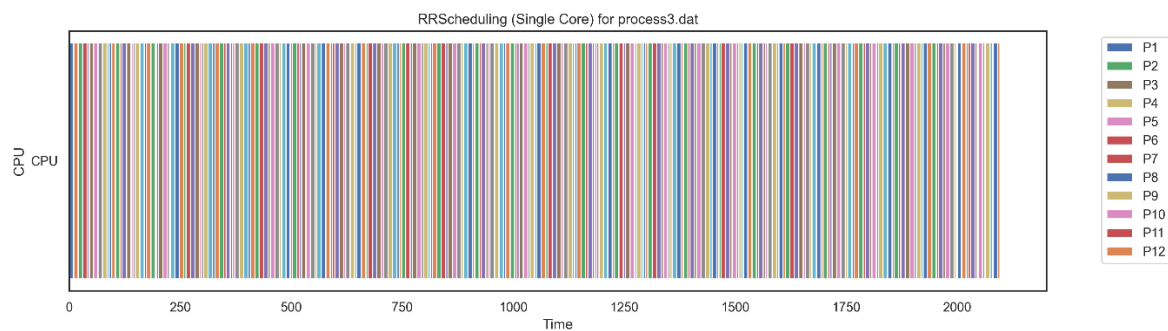
Time to Complete: 961

Maximum TAT: 935

Average TAT: 289.778

Runtime of Simulator: 959

Total runtime: 12308 microseconds



Process3.dat

Time to Complete: 2096

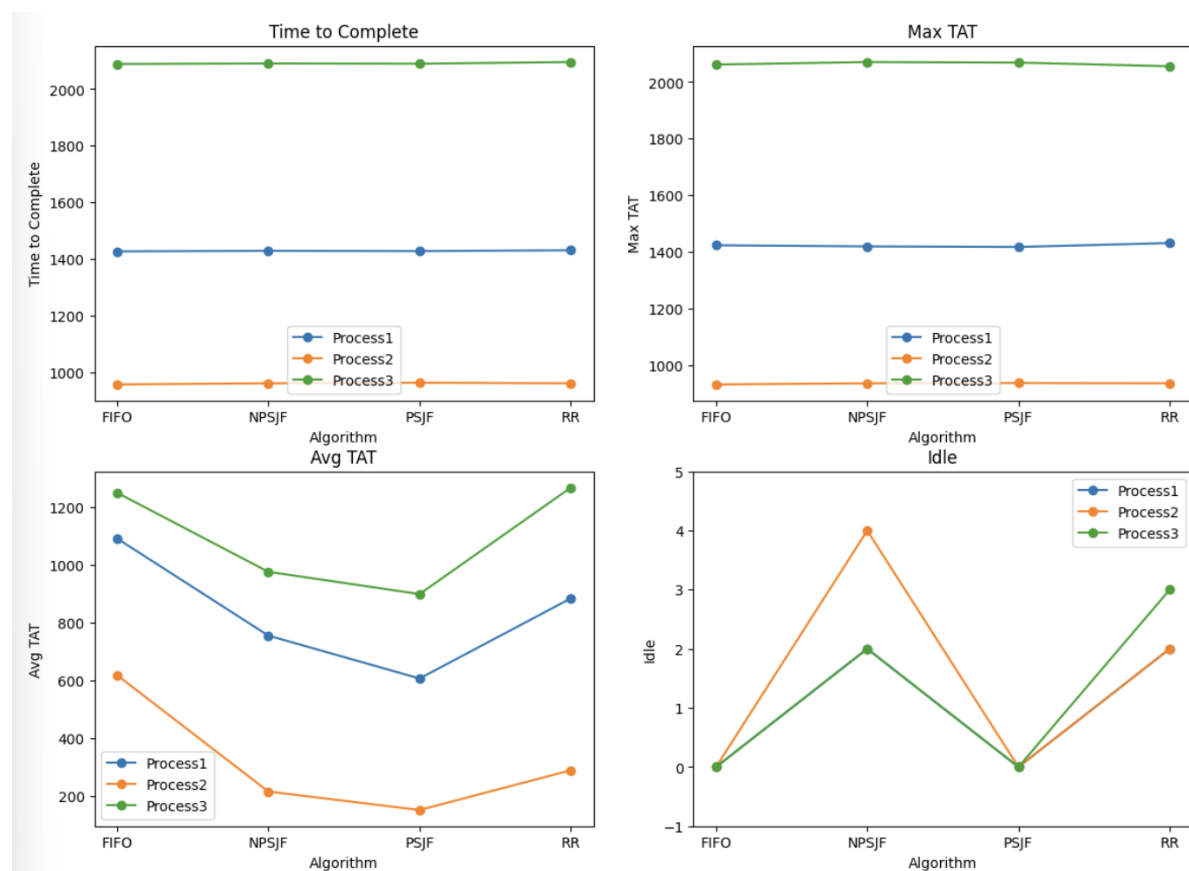
Maximum TAT: 2056

Average TAT: 1267

Runtime of Simulator: 2093

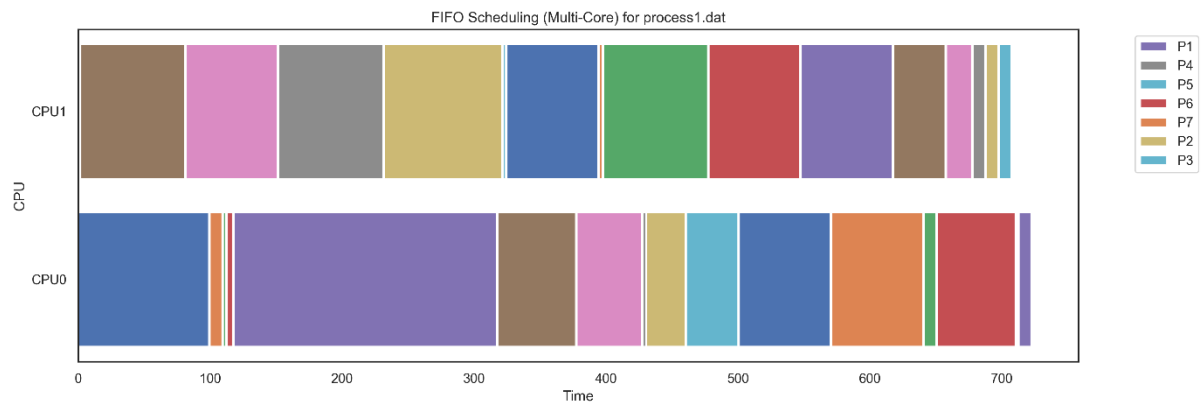
Total runtime: 33900 microseconds

Analysis for Single core Processor



2.MULTI CORE PROCESSOR

FIFO



Process1.dat

Time to Complete: 723

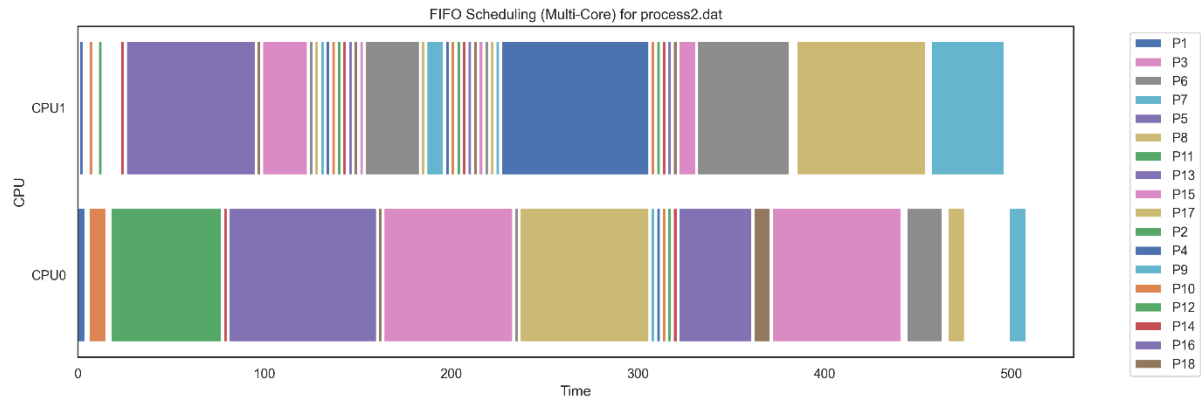
Maximum TAT: 723

Average TAT: 533.429

Runtime of Simulator(CPU1): 721

Runtime of Simulator(CPU2): 723

Total runtime: 20507 microseconds



Process2.dat

Time to Complete: 509

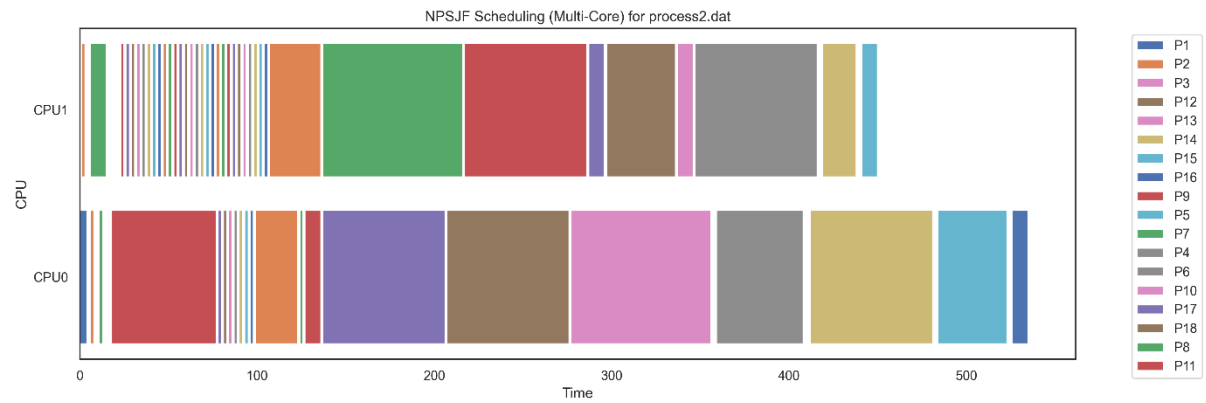
Maximum TAT: 483

Average TAT: 259.556

Runtime of Simulator(CPU1): 479

Runtime of Simulator(CPU2): 491

Total runtime: 21530 microseconds



Process2.dat

Time to complete: 536

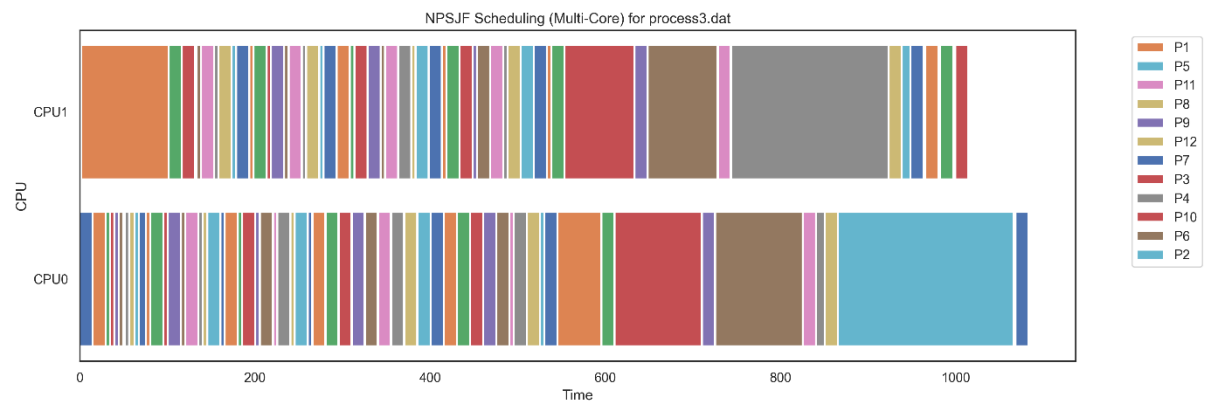
Average Turnaround Time: 109.556

Maximum Turnaround Time: 510

Runtime of Simulator(CPU1): 520

Runtime of Simulator(CPU2): 523

Total runtime: 19486 microseconds



Process3.dat

Time to complete: 1084

Average Turnaround Time: 531.167

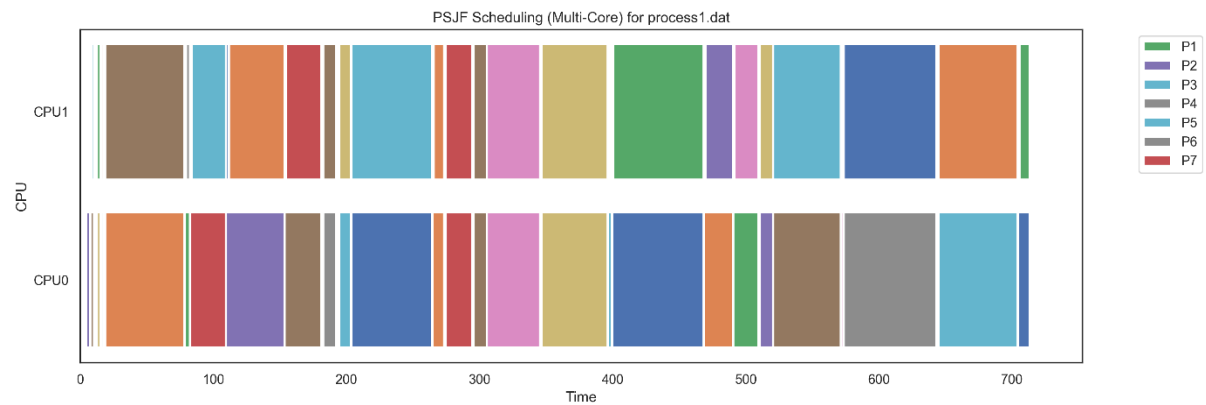
Maximum Turnaround Time: 1064

Runtime of Simulator(CPU1): 1082

Runtime of Simulator(CPU2): 1078

Total runtime: 24008 microseconds

PSJF



process1.dat

Simulation completed at time: 718

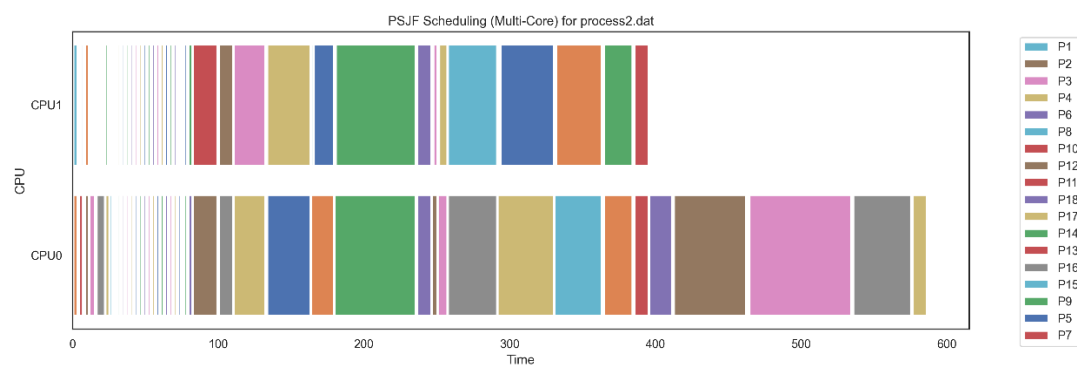
Average Turnaround Time: 351.714

Maximum Turnaround Time: 716

Runtime of Simulator(CPU1): 718

Runtime of Simulator(CPU2): 713

Total runtime: 27004 microseconds



process2.dat

Simulation completed at time: 587

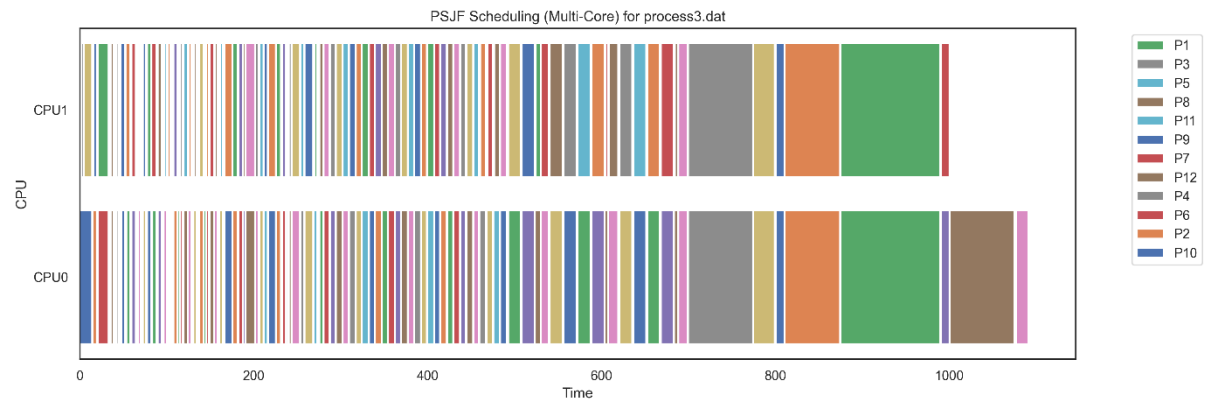
Average Turnaround Time: 87.0556

Maximum Turnaround Time: 561

Runtime of Simulator(CPU1): 581

Runtime of Simulator(CPU2): 568

Total runtime: 48739 microseconds



PSJF process3.dat

Simulation completed at time: 1092

Average Turnaround Time: 483

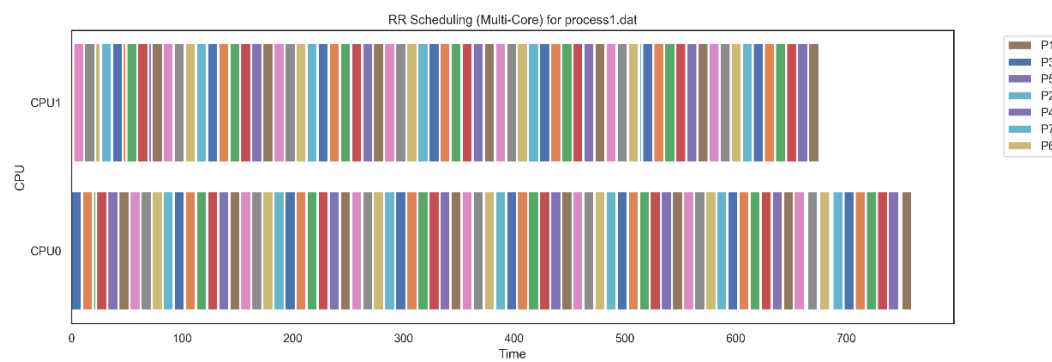
Maximum Turnaround Time: 1072

Runtime of Simulator(CPU1): 1091

Runtime of Simulator(CPU2): 1091

Total runtime: 101600 microseconds

RR



process1.dat

Time Slice: 10

Time to complete: 760

Average Turnaround Time: 445.714

Maximum Turnaround Time: 760

Runtime of Simulator(CPU1): 753

Runtime of Simulator(CPU2): 760

Total runtime: 17871 microseconds



process2.dat

Time Slice: 10

Time to complete: 526

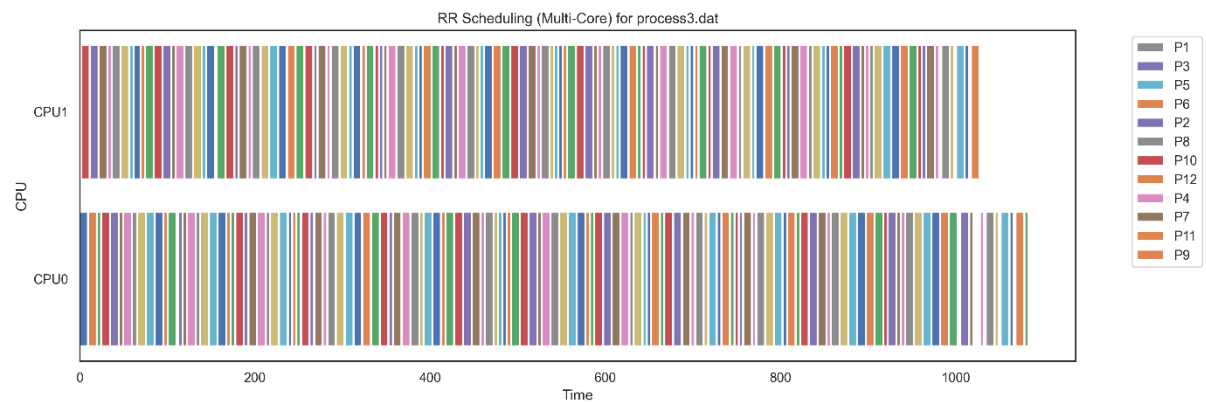
Average Turnaround Time: 139.389

Maximum Turnaround Time: 500

Runtime of Simulator(CPU1): 510

Runtime of Simulator(CPU2): 512

Total runtime: 11418 microseconds



process3.dat

Time Slice: 10

Time to complete: 1084

Average Turnaround Time: 628.75

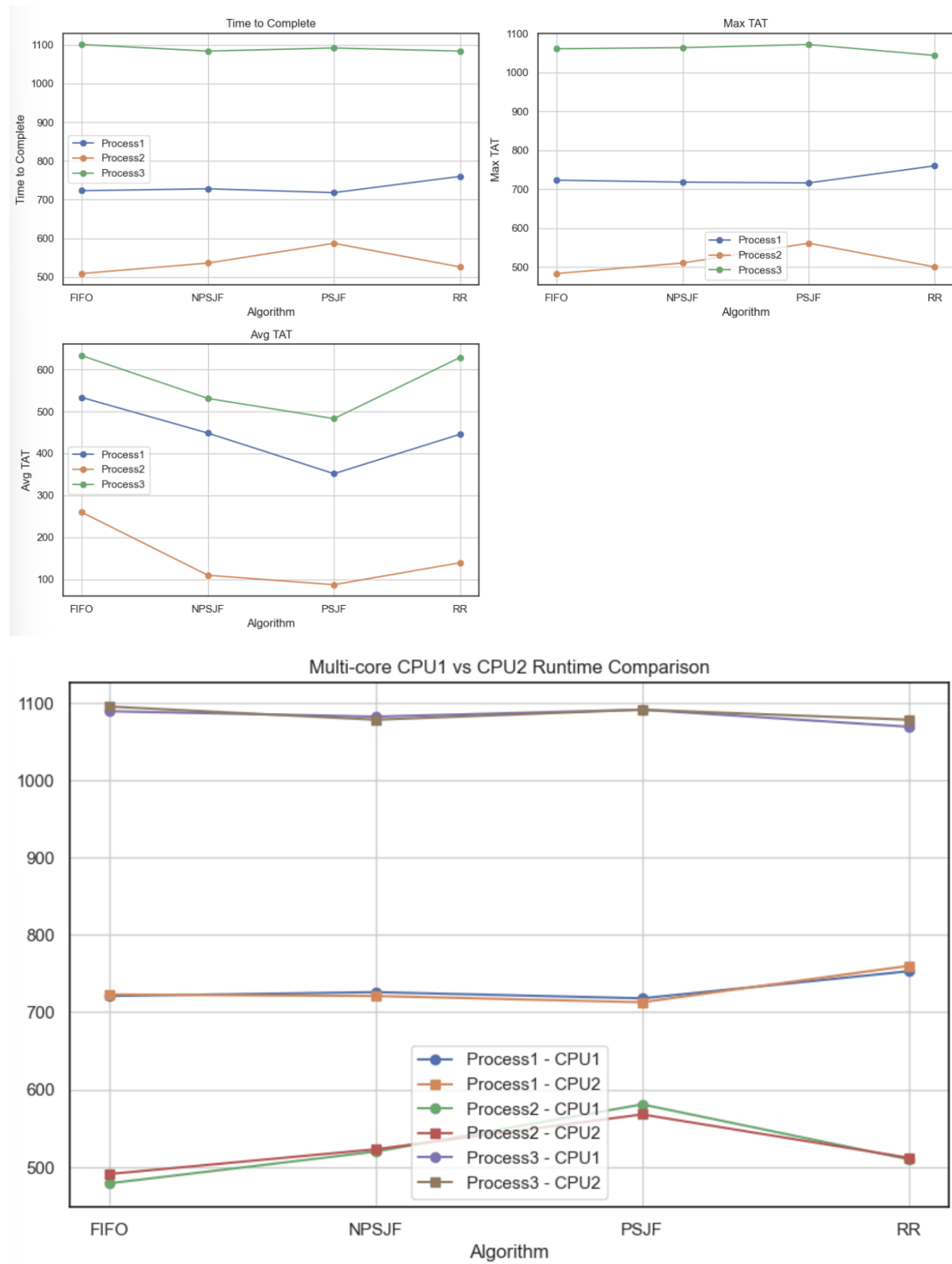
Maximum Turnaround Time: 1044

Runtime of Simulator(CPU1): 1069

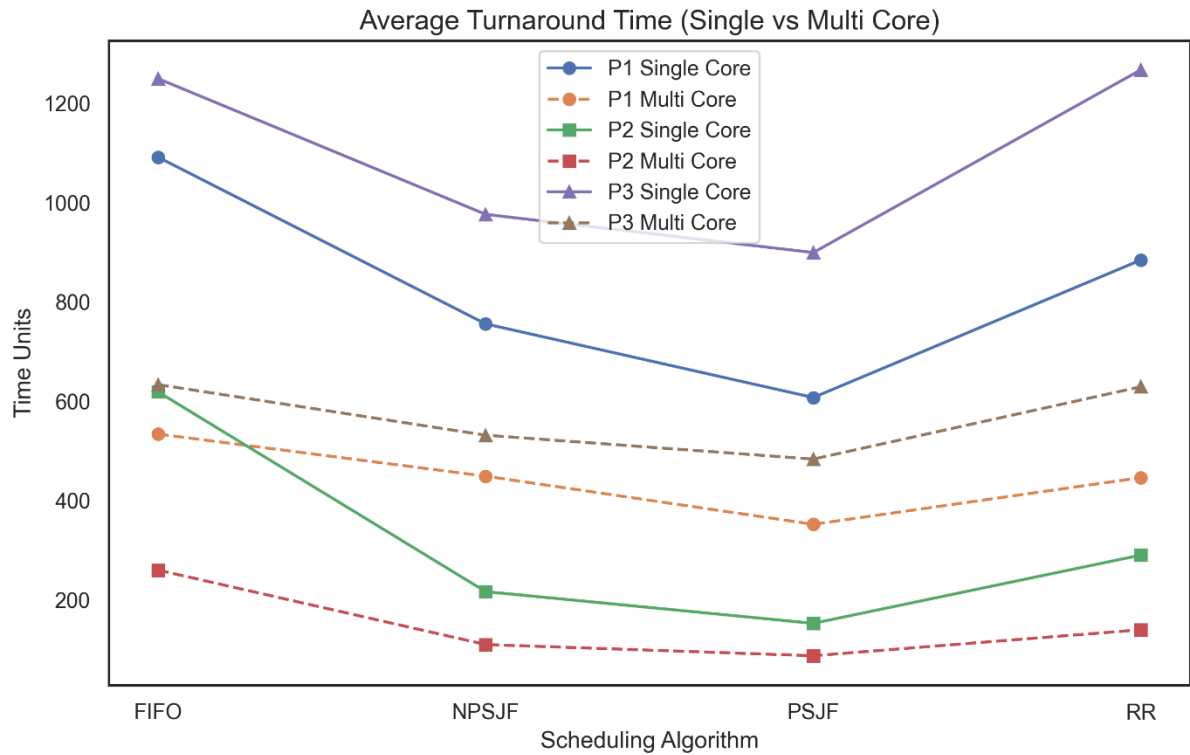
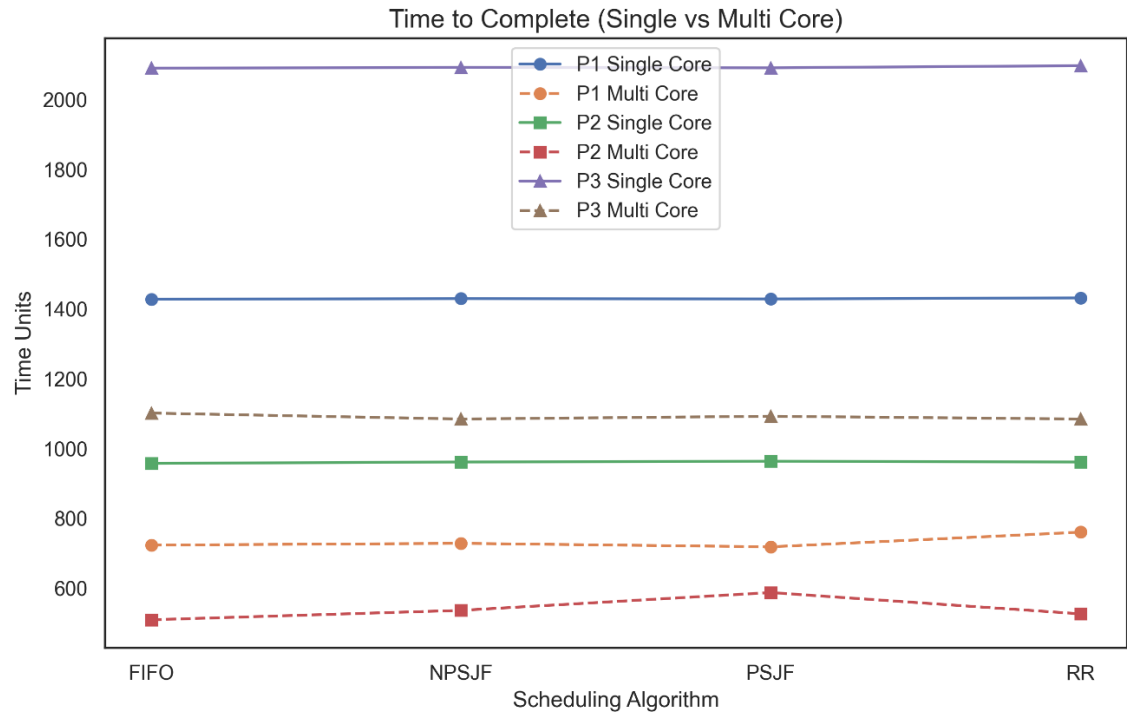
Runtime of Simulator(CPU2): 1078

Total runtime: 25937 microseconds

Analysis for Multi core Processor



COMPARATIVE ANALYSIS FOR SINGLE CORE AND MULTI CORE PROCESSOR



Conclusion

From the results, it is evident that multi-core processors perform significantly better than single-core processors in terms of turnaround time and overall runtime. Scheduling algorithms like PSJF and NPSJF achieve lower average turnaround times, while FIFO and RR show comparatively higher turnaround times. Overall, multi-core scheduling reduces execution time and improves efficiency, demonstrating the advantage of parallel execution over single-core processing.